EXERCISE – IMPLEMENTATION OF DECISION TREE

AIM

To implement a Decision Tree classifier using scikit-learn to classify Iris flower species based on sepal and petal measurements, and evaluate the model's accuracy.

PROCEDURE

1. Import the necessary libraries: scikit-learn for the Decision Tree classifier, dataset loading, train-test split, and accuracy metrics.
2. Load the Iris dataset, which contains 150 samples of three species of Iris flowers with four features each (sepal length, sepal width, petal length, petal width).
3. Separate the feature matrix (X) and the target labels (y).
4. Split the dataset into training and testing sets (typically 80% training and 20% testing).
5. Create and train a DecisionTreeClassifier model on the training data.
6. Use the trained model to predict the class labels for the test set.
7. Calculate and print the accuracy of the predictions compared to the true labels.
8. Run the program and observe the classification accuracy.

PROGRAM

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Train model
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

OUTPUT

```
Accuracy: 0.8666666666666667
```

Result: The Decision Tree classifier was successfully implemented and trained on the Iris dataset. The model achieved an accuracy of approximately 86.67% on the test set, demonstrating effective classification of Iris flower species based on the given features.